Guided Flow Matching for Extreme Weather Generation

Master's Thesis Presentation

Alain Joss
University of Zürich

Supervisors

Dr. Maxim Samarin **Dr. Shirin Goshtasbpour** **Prof. Dr. Guillaume Obozinski**
Swiss Data Science Center

Prof. Dr. Dan Olteanu
University of Zürich

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Motivation

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

Context:

- **Extreme weather events** play major role in society
- i.e. insurance, energy, agriculture, ...
- How to obtain **synthetic** scenarios?
- Idea: exploit generative weather models through ***guidance***

Guidance:

- Mature in image generation (e.g. Stable Diffusion [1])
- **Problem:** weather $\neq$ images

RQ:

Can we **steer** generative weather models toward prescribed **weather targets** while retaining a **structured atmospheric response**?

Flow matching

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

Flow matching [2] models a probability path $(p_t)_{t\in[0,1]}$:

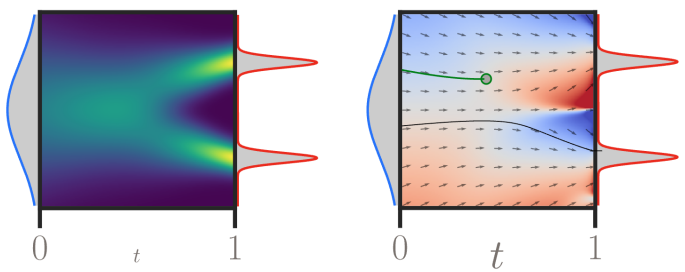
$$z_0 \sim p_0, \qquad z_t \sim p_t, \qquad z_1 \sim p_1$$

A convenient Gaussian path is

$$z_t = tz_1 + (1-t)\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I).$$

A neural network learns the corresponding velocity field:

$$\frac{\mathrm{d}}{\mathrm{d}t} z_t = u_t^\theta(z_t).$$



For this path, the model also provides a **clean-state estimate**

$$\boxed{\hat{z}_t = z_t + (1-t)u_t^\theta(z_t)}.$$

Guidance

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

Without guidance, integrating $u_t^\theta(z_t)$ generates samples from p_1.

Guidance: modify the velocity field toward a desired property y:

$$\boxed{u_t(z_t \mid y) = u_t(z_t) + c_t(y)}$$

Common structure:

$$\boxed{c_t(y) = w\,\gamma_t\,g_t(y)}$$

- $g_t(y)$: guidance direction
- $w \geq 0$: overall guidance gain
- $\gamma_t \geq 0$: guidance profile (relative stength) over flow time

Constructing the guidance intervention

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

1. Local loss-based guidance [3]

$$g_t(y) = -\nabla_{z_t} \mathcal{L}(\hat{z}_t, y)$$

2. Control-based guidance [4]

$$c^{\star}_{0:1} = \arg\min_{c_{0:1}} \left[\mathcal{L}(z_1, y) + \eta \int_0^1 C(c_t)\, \mathrm{d}t \right]$$

Weather applications:

- tropical-cyclone generation [5]
- precipitation interventions [6]

Open questions $(w\, \gamma_t g_t)$

- How to adapt w to **realize the prescribed target** y?
- How to set γ_t to obtain the most **structured atmospheric response**?

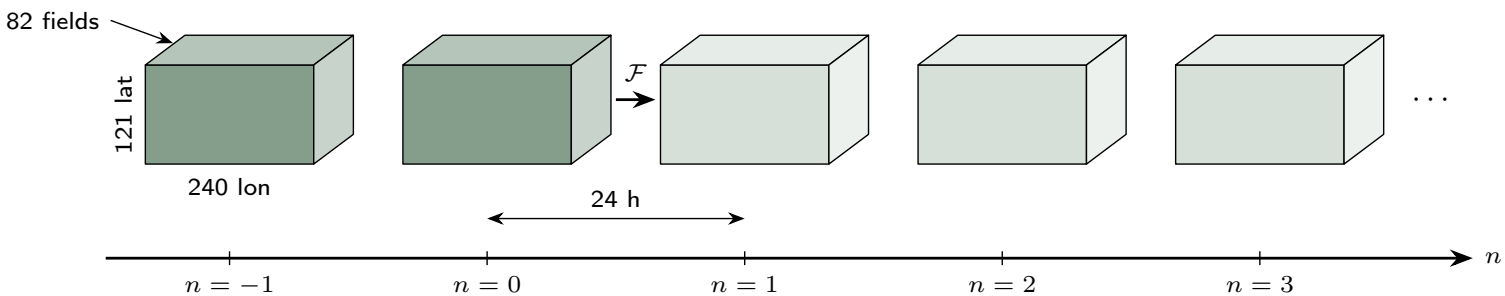
ArchesWeather

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

We use **ArchesWeather** [7] as generative model:

$$\hat{x}_n = \mathcal{F}\left(x_{n-2}, x_{n-1}, \text{seed}\right)$$

trained on a version ERA5 with $x^{82\times121\times240}$.



Scenario specification

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

Scenario specification (y):

$$\boxed{(v^\star,\, \ell^\star,\, \pi,\, \rho)}$$

Target variable $v^\star$ and level $\ell^\star$

$$v^\star = \mathrm{t2m}, \qquad \ell^\star = 0$$

Spatial average $\mathcal{M}_\pi$

$$\mathcal{M}_\pi(x_{v\ell}) := \sum_{i,j} \pi_{ij} x_{v\ell ij}, \quad \sum_{i,j} \pi_{ij} = 1$$

Target region π



Targets with relative intensity ρ

$$\boxed{y_{n,m} := (1+\rho)\, \mathcal{M}_\pi \left(x^{\mathrm{ung}}_{n,m,v^\star \ell^\star}\right)}$$

for each ensemble member m

Guidance method

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

Recall

$$\boxed{c_t = w\,\gamma_t\,g_t}$$

Guidance direction

$$\mathcal{L}_t := \xi_t^2 = \left[\mathcal{M}_\pi\left(\hat{x}_{t,v^\star\ell^\star}\right) - y\right]^2$$

$$\boxed{g_t := -\nabla_{z_t}\mathcal{L}_t}$$

Gain adaptation

For a fixed profile γ_t, adapt w s.t.

$$\boxed{\xi_1(w^\star) = 0} \qquad w^\star \longleftarrow \text{secant search}$$

Experimental design

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

Goal: isolate the effect of guidance timing $(w\,\boldsymbol{\gamma}_t\,g_t)$

Profiles

$$\text{spike@}k: \qquad \gamma_s = \mathbf{1}_{\{s=k\}}, \qquad k \in \{0, 1, 2, 5, 11, 17, 23\}.$$

Matched comparison:
Across profiles: same input states, seed, same achieved target
$\rightarrow$ isolate guidance effect:

$$\boxed{\Delta x^{\mathrm{GE}} := x^{\mathrm{gui}} - x^{\mathrm{ung}}}$$

Experimental setup

Initializations	Intensity	Ensemble	Flow steps
4 dates	$\rho \in \{0.002, 0.004\}$	$M = 5$	$K = 25$

Evaluation:

- guidance direction g_s and guidance effect Δx_s^{GE}
- multivariate propagation of Δx_s^{GE}

g_s and Δx_s^{GE}

Introduction

Background

Methodology

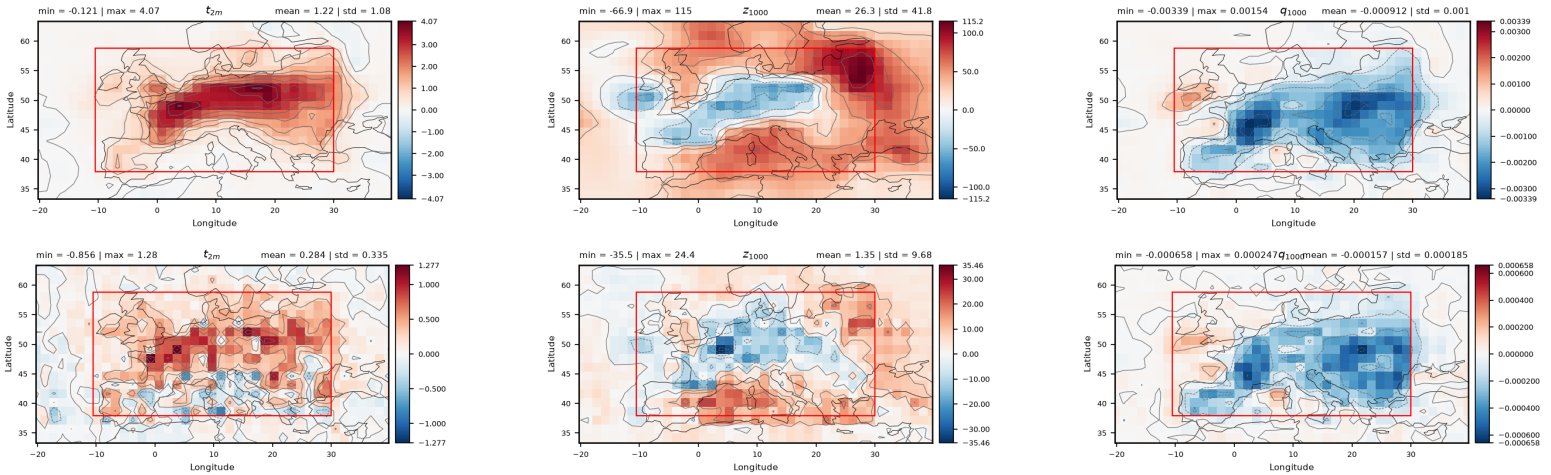
Experimental evaluation

Discussion

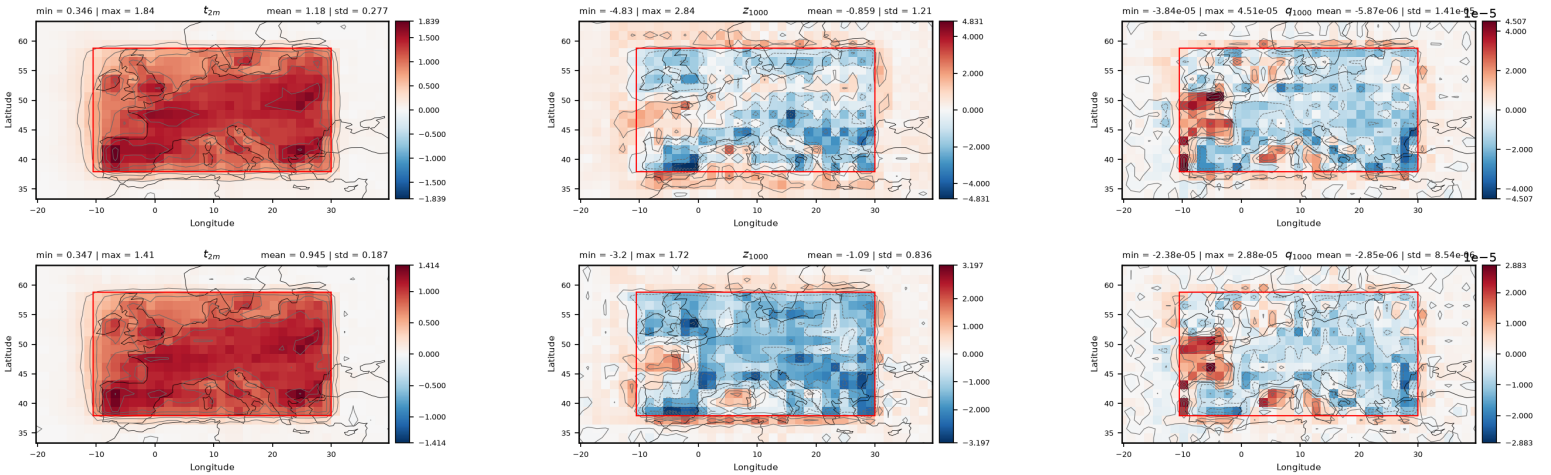
References

2026-07-09, $\rho = 0.004$ top: Δx_s^{GE}, bottom: g_s

Early guidance: spike@1



Late guidance: spike@17



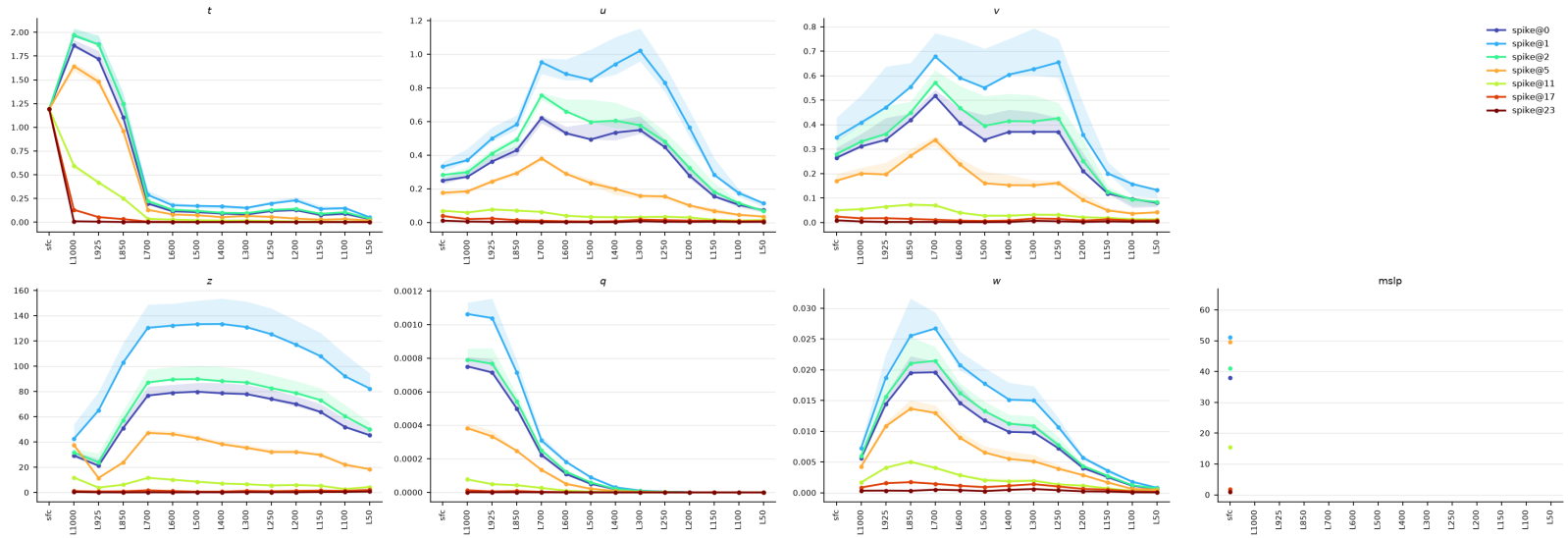
Multivariate propagation

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

For each variable–level pair (v, ℓ):

$$P_{v\ell}^{\gamma} := \mathcal{M}_{\pi}\left(\left|\Delta x_{v\ell}^{\mathrm{GE},\gamma}\right|\right)$$

2026-07-09, $\rho = 0.004$



earlier guidance $\longrightarrow$ stronger multivariate propagation

Which guidance profile performs best?

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

Across experiments:

$$P^{\gamma}_{v\ell}$$

Profile	Multivariate Propagation
spike@0	0.689 ± 0.050
spike@1	$\mathbf{1.000 \pm 0.000}$
spike@2	0.770 ± 0.057
spike@5	0.463 ± 0.040
spike@11	0.134 ± 0.013
spike@17	0.046 ± 0.004
spike@23	0.028 ± 0.002

Mean $\pm$ population std; normalized per setting.

spike@1 shows the strongest multivariate propagation

Interpreting the timing effect

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

Guidance signal

$$g_t = \nabla_{z_t} \mathcal{L}_t$$

- depends on z_t
- depends on t
- increasingly target-localized

Remaining flow

$$z_t \longrightarrow z_1$$

- propagates the intervention
- variables coupling
- smoothing

Interpretation

early guidance: less localized signal + more remaining flow

Conclusion

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

Can we steer generative weather models toward prescribed targets while retaining a structured atmospheric response?

- **Target control:** adaptive guidance realizes the prescribed regional target.
- **Structured response:**
  $\rightarrow$ among the tested profiles, early guidance produces
  - least pathological spatial response
  - strongest multivariate propagation

$$\boxed{\gamma = \text{spike@1}}$$

Future research:

$$\boxed{\text{structured response} \not\Rightarrow \text{meteorological plausibility}}$$

References I

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

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References II

Introduction
Background
Methodology
Experimental evaluation
Discussion
References

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